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A multi-agent large language model workflow for analyzing perceived cultural values from social media: A study of 141 Chinese cities

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ABSTRACT

Heritage attractions are key urban assets that embody cultural values, shape local identity, and support economic development through tourism. Understanding the values the public perceives in these assets, and the reasons behind these perceptions from a bottom-up, public-centered perspective, is essential for evidence-based urban and heritage policy. Social media data (SMD) presents opportunities for such analysis, yet current methods struggle to scale to national coverage and to convert unstructured text into operational, decision-ready urban knowledge. To address these challenges, we introduce a multi-agent large language model workflow that identifies and quantifies eight cultural values from SMD and extracts supporting evidence to construct Cultural Heritage Perception Graphs (CHPGs), which structure and reveal the underlying reasons. Analyzing 284,030 online comments on 703 heritage attractions across 141 Chinese cities, we map how different cultural values are expressed, co-occur, and vary spatially across cities. The results highlight the prominence of aesthetical and social dimensions in public discourse, alongside more selective and context-dependent expressions of the other value dimensions. At the city level, overall perception scores correlate positively with comment volume, tourism revenue, and GDP per capita, revealing four distinct city profiles that can guide targeted management strategies. In practice, our CHPG-powered retrieval-augmented generation (RAG) system translates unstructured SMD into interactive, evidence-based insights, helping to bridge the gap between public feedback and actionable decision-making and supporting heritage management that is more responsive to public perception.

1. Introduction

Cultural heritage represents the living legacy of a society's values, beliefs, knowledge, and traditions, embodied in both tangible and intangible forms (Council of Europe, 2005). Within urban contexts, heritage functions not only as a cultural asset but also as a driver of economic vitality, as heritage attractions, including sites, monuments, and historic districts contribute to urban tourism through visitor spending, employment, and destination branding (Guzmán et al., 2017). Given this significance, the critical task extends beyond merely protecting historical spaces to actively revitalizing these resources by deepening our understanding of heritage values and integrating this knowledge into management and policy frameworks (Avrami et al., 2019).

A critical paradigm shift is underway in understanding heritage values, moving from expert-driven, top-down assessments toward public-centered, bottom-up interpretations that recognize heritage as a socio-cultural construct co-created by diverse communities (Bai, 2023; Ginzarly et al., 2019; Ginzarly & Srour, 2022; Havinga et al., 2020; Jones, 2017; Veldpaus, 2015). Supported by UNESCO's Recommendation on the Historic Urban Landscape (HUL) (UNESCO, 2011), this shift calls for integrating perspectives from tourists, residents, and other stakeholders into heritage value assessment. While traditional methods such as surveys and interviews are often constrained by high costs, long timeframes, potential biases, and limited scale (Ginzarly et al., 2019; Luo et al., 2025; Zhao et al., 2024; Zhao, Huang, Lin, & Lu, 2025), social media data (SMD) has emerged as a promising and cost-effective alternative for capturing public perspectives at scale (Bai, Nourian, & Roders, 2021; Ginzarly et al., 2019; Ginzarly & Srour, 2022). However, two key gaps continue to limit the use of SMD for public-centered heritage management.

First, existing empirical studies on public-perceived cultural values

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have been limited to individual heritage sites or single cities (Bai et al., 2023; Foroughi et al., 2023; Ginzarly & Srour, 2022; Koufodontis & Gaki, 2022). While these case studies offer rich, context-specific insights, they cannot reveal how perceived cultural values are distributed and interrelated at a national scale and what common patterns characterize public discourse on different value dimensions. Yet such cross-city, large-scale understanding is what policymakers need for formulating differentiated management strategies and identifying cities whose cultural assets are underrecognized or underutilized (Guzmán et al., 2017; Havinga et al., 2020). This gap is largely attributable to methodological constraints: existing value identification approaches rely on lexicon matching or text classification (Bai et al., 2022; Bai et al., 2023; Boota et al., 2025; Fan & Chen, 2024; Foroughi et al., 2023; Koufodontis & Gaki, 2022) (Table S1), which require labor-intensive construction of value-specific dictionaries or annotated training data and are difficult to generalize across diverse linguistic and cultural contexts.

Second, beyond identifying what values the public perceives, understanding what generates those perceptions is equally important for guiding management interventions, particularly for addressing negatively perceived aspects (Ginzarly et al., 2019; Havinga et al., 2020; Silva & Roders, 2012; UNESCO, 2011; Veldpaus, 2015). Yet even when values are successfully identified, the analytical outputs that could inform such understanding typically remain fragmented as isolated keywords or topic labels lacking organization and interconnection (Foroughi et al., 2023; Gao et al., 2022; Ginzarly et al., 2019). For instance, word frequency analysis may reveal that "beautiful" is a high-frequency term in visitor comments about the Palace Museum, but it cannot specify whether visitors are frequently referring to the architecture or the gardens; conversely, given a specific heritage feature, existing methods cannot readily aggregate what visitors praise or criticize about it, or how these perceptions connect to those of other features. These fragmented outputs make it difficult to retrieve complete evidence chains and trace interconnected factors contributing to value perceptions. Therefore, a key challenge is not only to detect values from social media text, but also to transform value-related text into structured and queryable knowledge that can support evidence-based decision-making (Bai, Luo, et al., 2021; Foroughi et al., 2023; Ginzarly et al., 2019).

Recent advances in large language models (LLMs) and knowledge graphs (KGs) offer promising solutions. LLMs across the GPT, Gemini, DeepSeek, and GLM families demonstrate human-like capabilities in language understanding, complex reasoning, and information extraction (Kaur et al., 2024; Xi et al., 2025). Their few-shot and zero-shot learning capabilities reduce reliance on manually labeled data, enabling more scalable value identification and value-related evidence extraction directly from SMD (Luo et al., 2025; Zhao, Huang, Yang, et al., 2025). For structuring these extracted insights, KGs organize knowledge as networks of interconnected triples, each linking two entities through a relationship. Because entities are shared across triples, a KG naturally chains fragmented information into structured, machine-readable knowledge (Fan & Chen, 2024; Lu et al., 2026; Ning & Liu, 2024; Wang, Liu, et al., 2024). For example, the triple <the Great Wall, evokes, sense of national pride> captures a specific value perception, and through the shared entity "the Great Wall," connects to other triples describing its attributes and related context, enabling traversal from perceived values to their underlying evidence. Together, LLMs extract insights from SMD, while KGs structure them into interconnected knowledge that links perceived values to the heritage experiences generating them. Yet such integrated applications in heritage research remain scarce.

Responding to UNESCO's HUL call for civic engagement tools that support stakeholder participation in value identification (UNESCO, 2011), we propose a multi-agent LLM workflow that identifies public-perceived cultural values from SMD following Roders (2007)'s eight-value typology and extracts the supporting evidence as structured entity-relationship triples. These triples are organized through a custom Cultural Heritage Perception Ontology and compiled into queryable Cultural Heritage Perception Graphs, which can power downstream applications such as retrieval-augmented generation for evidence-based querying. We apply this workflow to 284,030 online comments on 703 heritage attractions across 141 Chinese cities, mapping how different cultural values are distributed spatially, how they relate to urban characteristics, and what specific aspects of the experience generate public perceptions of each value dimension, aiming to answer:

RQ1. How can perceived cultural values and their corresponding value evidence be scalably identified, measured, structured, and operationalized from large-scale SMD?

RQ2. What are the national-scale distribution patterns of perceived cultural values across China, how are these perceptions interrelated and connected to urban characteristics, and what value evidence underlies public discourse on different value dimensions?

2. Literature review

2.1. Cultural values in social media research: Frameworks and directions

Cultural values are commonly defined as "the different qualities, characteristics, meanings, perceptions, or associations ascribed to the things we wish to conserve" (Avrami et al., 2019). Over decades, various international charters, from the Venice Charter (1960s) to the World Heritage Convention (1970s), the Washington and Burra Charters (1980s–1990s), and more recent guidelines such as the Valletta Principles and the HUL Recommendation, have proposed diverse value systems. Veldpaus (2015) consolidated these frameworks and argued that they can be grouped into the eight cultural values defined by Roders (2007): social, economic, political, historic, aesthetical, scientific, age, and ecological. These categories form a comprehensive framework capturing both tangible and intangible heritage significance and have been widely adopted in heritage research (Foroughi et al., 2023; Havinga et al., 2020; Silva & Roders, 2012).

Within this values-based perspective, heritage assessment has long been understood as linking values to the attributes that carry them (Ginzarly et al., 2019; Havinga et al., 2020; Silva & Roders, 2012; UNESCO, 2011; Veldpaus, 2015). For example, Fredheim and Khalaf (2016) argued that heritage assessment should address what heritage is, why it is valuable, and how valuable it is. This logic has also been extended to SMD-based studies. Ginzarly et al. (2019) used Flickr metadata in Tripoli to relate viewpoint locations, depicted scenes, and tags to perceived values and heritage attributes, showing that public understandings often extend beyond formally designated areas and are articulated through everyday vocabularies. Foroughi et al. (2023) analyzed 23,899 Instagram and Twitter posts on Yazd's windcatchers based on Roders (2007)'s value typology, finding that age, historic, and social values were most prominent and that value-related evidence was often tied to specific attribute types such as buildings, building elements, and natural elements. Bai et al. (2023) used multimodal Flickr data from Venice to classify and spatially map perceived cultural significance, revealing associations between value-related and attribute-related categories, for example between urban form elements and categories related to testimony, typology, and association.

These studies show that SMD can support public-centered heritage assessment by revealing both perceived values and the specific attributes associated with them. However, most studies focus on a single site or city, and large-scale investigations of how different cultural values are perceived remain absent. Koufodontis and Gaki (2022) broadened the scope to 105 World Heritage Sites by using more than 230 predefined expressions to detect tourist awareness of UNESCO designation in TripAdvisor reviews. However, building exhaustive lexicons for fine-grained cultural value identification at a large scale is impractical. Furthermore, existing studies have primarily linked values to heritage attributes. Yet social media discourse reveals that value perceptions are also shaped by contextual factors such as visitor facilities, organized

events, and historical narratives, which have rarely been extracted and organized. Capturing these broader factors, which we refer to as "value evidence" in this study, requires moving beyond attribute-focused frameworks toward structured knowledge representations that can accommodate the full range of entities and relationships present in public heritage discourse.

2.2. Knowledge graphs in cultural heritage: Current applications and gaps

KGs have emerged as a powerful approach for organizing and representing knowledge about both tangible and intangible heritage (Carriero et al., 2019; Fan & Chen, 2024; Fan & Wang, 2022;Pellegrino et al., 2022; Wang, Liu, et al., 2024). By structuring data from diverse sources, KGs enable applications such as semantic search and complex querying (Carriero et al., 2019; Fan & Chen, 2024; Pellegrino et al., 2022), thereby improving information retrieval and supporting heritage management decision-making (Fan & Chen, 2024; Wang, Liu, et al., 2024).

These KGs are typically built upon ontological frameworks designed for expert documentation and populated with formal, expert-curated sources such as scholarly articles, official archives (Pellegrino et al., 2022; Wang, Liu, et al., 2024), and geospatial databases (Ning & Liu, 2024). CIDOC-CRM, the most widely adopted ontology for heritage documentation, models entities such as physical objects, actors, events, and temporal spans to represent provenance, conservation history, and material properties (Doerr, 2003), while domain-specific extensions such as ArCo (Carriero et al., 2019) further standardize expert terminology. While effective for formal heritage records, these ontologies were not designed to accommodate the entity types that emerge from public discourse on social media, leaving an ontological gap between expert-oriented knowledge structures and the representation of public heritage perceptions.

Moreover, the inherent characteristics of SMD, its sheer volume, noise, and linguistic informality (e.g., slang, abbreviations (Boota et al., 2025)), pose challenges for conventional KG construction. Traditional techniques, such as rule-based methods (Ning & Liu, 2024) or standard named entity recognition models (Fan & Wang, 2022; Gao et al., 2022; Wang, Liu, et al., 2024), often struggle to accurately extract entities and their relationships from such messy data. Recent research has begun to address these issues by integrating LLMs into KG construction workflows, leveraging their strong capabilities for autonomous, domain-specific information extraction from complex texts (Lu et al., 2026; Ning & Liu, 2024). However, the application of LLMs for constructing KGs specifically from SMD remains in its early stages.

2.3. Evolving solutions: From single LLM to multi-agent LLM workflow

While LLMs show significant promise for processing SMD (Li et al., 2025; Luo et al., 2025), their limitations include a tendency to "hallucinate" by generating fluent yet factually incorrect outputs, as well as low reliability in complex, multi-step tasks (Zhao, Huang, Yang, et al., 2025). To mitigate these issues, researchers have increasingly turned to LLM-based multi-agent systems. Inspired by the concept that intelligence emerges from the interaction of specialized agents (Minsky, 1988), these systems employ multiple autonomous LLM-based agents that collaborate to solve complex problems, enhancing the overall reliability, scalability, and adaptability of the outputs (Ning & Liu, 2024; Xi et al., 2025; Zhou et al., 2024).

This approach of assigning specialized roles to foster collaboration has been successfully applied in diverse fields, including software development (Chen et al., 2023; Hong et al., 2023), robotics (Kannan et al., 2024), and patient diagnosis (Li et al., 2024). Particularly relevant to this study, agent-based frameworks in urban studies have leveraged LLMs' human-like reasoning and role-playing capabilities to simulate complex social dynamics such as urban mobility (Shao et al., 2024; Yan et al., 2024). This multi-agent paradigm is particularly well-suited to our objectives of analyzing the cultural values expressed in SMD.

3. Materials and methods

3.1. Theoretical framework and research workflow

We conceptualize public-centered heritage assessment as a two-part process: evaluating the cultural values perceived by the public and organizing the value evidence that generates these perceptions (Fig. 1). Roders (2007)'s eight-value typology (social, economic, political, historic, aesthetical, scientific, age, ecological) provides the conceptual foundation for identifying and quantifying cultural values in SMD. Definitions of each value and its sub-types are provided in *Appendix A*. To explain why these values are perceived, the extracted value evidence is structured into Cultural Heritage Perception Graphs (CHPGs) using a Cultural Heritage Perception Ontology (CHPO) comprising six entity types: Objects, Actors, Events, Time, Place, and Characteristics. This ontology links values to their influencing factors and organizes SMD evidence into an interpretable and queryable form.

As illustrated in Fig. 2, our research involved four stages: data collection, multi-agent processing of SMD, data analysis, and the application of a CHPG-powered RAG system.

3.2. Data collection and preprocessing

Our study focuses on China's 143 National Historical and Cultural Cities, designated by the Chinese government for the preservation of urban areas with profound cultural and historical significance. From these cities, we selected the top five prominent heritage attractions in each city based on total comment volume on Ctrip (www.ctrip.com), one of China's largest online travel agencies. Prior studies have validated Ctrip SMD as a reliable source for urban research (Luo et al., 2025; Zhao et al., 2024; Zhao, Huang, Lin, & Lu, 2025).

We collected comments posted between January 1, 2024, and December 31, 2024. For attractions with under 50 comments in 2024, we supplemented with all available historical comments. To ensure data quality, attractions with fewer than 20 historical comments were excluded. This process yielded 310,971 raw comments from 703 attractions across 141 cities. Our coverage analysis shows that the top five attractions account for an average of 84.81% of all heritage-related comments per city, indicating that our sample captures the dominant online discourse about heritage in each city. After cleaning to remove duplicate, promotional, and irrelevant content, the final dataset comprised 284,030 valid comments totaling 12,556,007 characters, with an average length of 43.87 characters and a maximum of 1129 characters.

3.3. Multi-agent LLM workflow

Our multi-agent LLM workflow was built on the MetaGPT framework (https://github.com/geekan/MetaGPT) (Hong et al., 2023) and is illustrated in Fig. 3. Detailed prompts for each agent are provided in *Appendix B*.

3.3.1. Value identification and initial evidence extraction

1) **Pre-Identification Agent:** This agent analyzes a comment C to identify which of the eight cultural values V are present. For each identified value $v_i \in V$, it extracts the value-related text chunks T_{v_i}. This process can be modeled as:

$$f_{pre}(C) \rightarrow \left(V_C \subseteq V, \{T_{v_i}\}_{v_i \in V_C} \right) \quad (1)$$

2) **Verification Agent:** It performs quality control to improve the reliability of the initial extraction. It takes the original comment C and

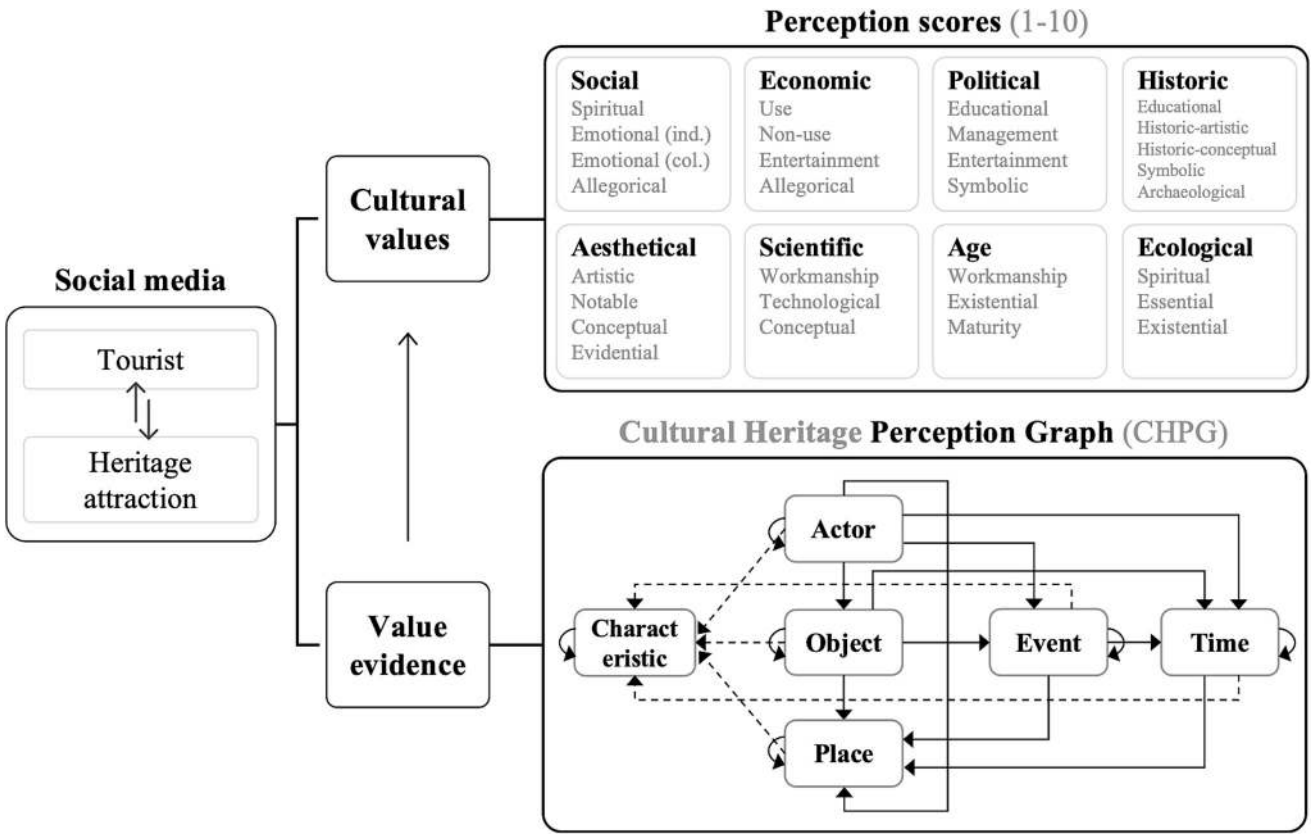


Fig. 1. Theoretical framework for assessing cultural values from SMD.

the pre-identified values and text chunks ($V_C, \{T_{v_i}\}$) as input for a second assessment. This agent verifies the accuracy of the assigned values and refines the extracted text chunks for conciseness and alignment. This yields a verified set of values $V_C^{'}$ and value-related text chunks $\left\{T_{v_i}^{'}\right\}$, ensuring high-fidelity input for subsequent stages. The function is:

$$\mathrm{f_{ver}}\left(\mathrm{C}, \mathrm{V_C}, \{T_{v_i}\}\right) \rightarrow \left(V_C^{'}, \left\{T_{v_i}^{'}\right\}_{v \in V_C^{'}}\right) \tag{2}$$

3) **Dimension-Specific Analysis Agents:** The verified data is then sent to eight specialized agents, each an expert in one value dimension. An agent is only activated if its value was verified in the previous step. Activated agents perform two parallel tasks on their assigned text chunks $T_{v_i}^{'}$:

Perception scoring: Each agent assesses the emotional tone of the texts to assign a perception score S_{v_i} on an integral scale from 1 to 10. This is modeled as:

$$S_{v_i} = \sigma\left(T_{v_i}^{'}, C\right) \in [1, 10] \tag{3}$$

Knowledge triple extraction: Simultaneously, each agent converts the unstructured text chunks into a structured format by identifying knowledge triples in the form <Entity/Subject, Relation, Entity/Object>. This process, $f_{extract}$, captures key entities and relationships within the text chunks:

$$Triple_{v_i} = f_{extract}\left(T_{v_i}^{'}, C\right) = \left\{\left(E_j^{\mathrm{sub}}, R_j, \mathrm{E_j^{obj}}\right)\right\}_{j=1}^{k} \tag{4}$$

4) **Evaluation protocol:** To evaluate this fully-automated stage, eight LLMs were benchmarked on a manually curated test dataset of 1000 high-quality comments. The models tested were Gemini-2.5-Flash, GPT-4.1-mini, DeepSeek-V3, GLM-4-Plus, GLM-4-Air, Hunyuan-T1, Qwen3-32b, and Qwen-Turbo. For perception scoring, performance was measured against human-annotated ground truth using Mean Absolute Error (MAE) and Mean Squared Error (MSE). For value-related triple extraction, dimensional relevance, information completeness, and content accuracy were evaluated. The scale of this evaluation involved 192,000 individual assessments (1000 comments × 8 models × 8 values × 3 dimensions), which made exhaustive manual annotation impractical. Moreover, as described in *Section 3.3.2*, the subsequent triple postprocessing stages incorporate extensive manual verification and prompt revision, providing an additional layer of human quality assurance for the final outputs. Therefore, we employed an LLM-as-judge protocol (Lu et al., 2026; Ning & Liu, 2024) to assess the models' comparative capabilities at this initial extraction stage. In this protocol, each LLM cross-evaluated the outputs from the other models. The final score for each model was the average of all judgments.

3.3.2. Triple refinement through iterative human-agent collaboration

During our experiments, we observed that the large volume of Chinese comments exhibits highly diverse and often irregular linguistic patterns, and applying fixed prompts introduced various quality issues in the extracted triples. To ensure the quality of the final triple lists, the following agents were not applied as one-time automated processes but through iterative cycles of human-agent collaboration. In each cycle, we systematically reviewed the agent's outputs to identify recurring errors, then carefully revised the prompts with targeted few-shot examples to guide the agent in rectifying these issues, repeating until the outputs reached satisfactory quality. These iterative stages are marked “multiple rounds” in Fig. 3.

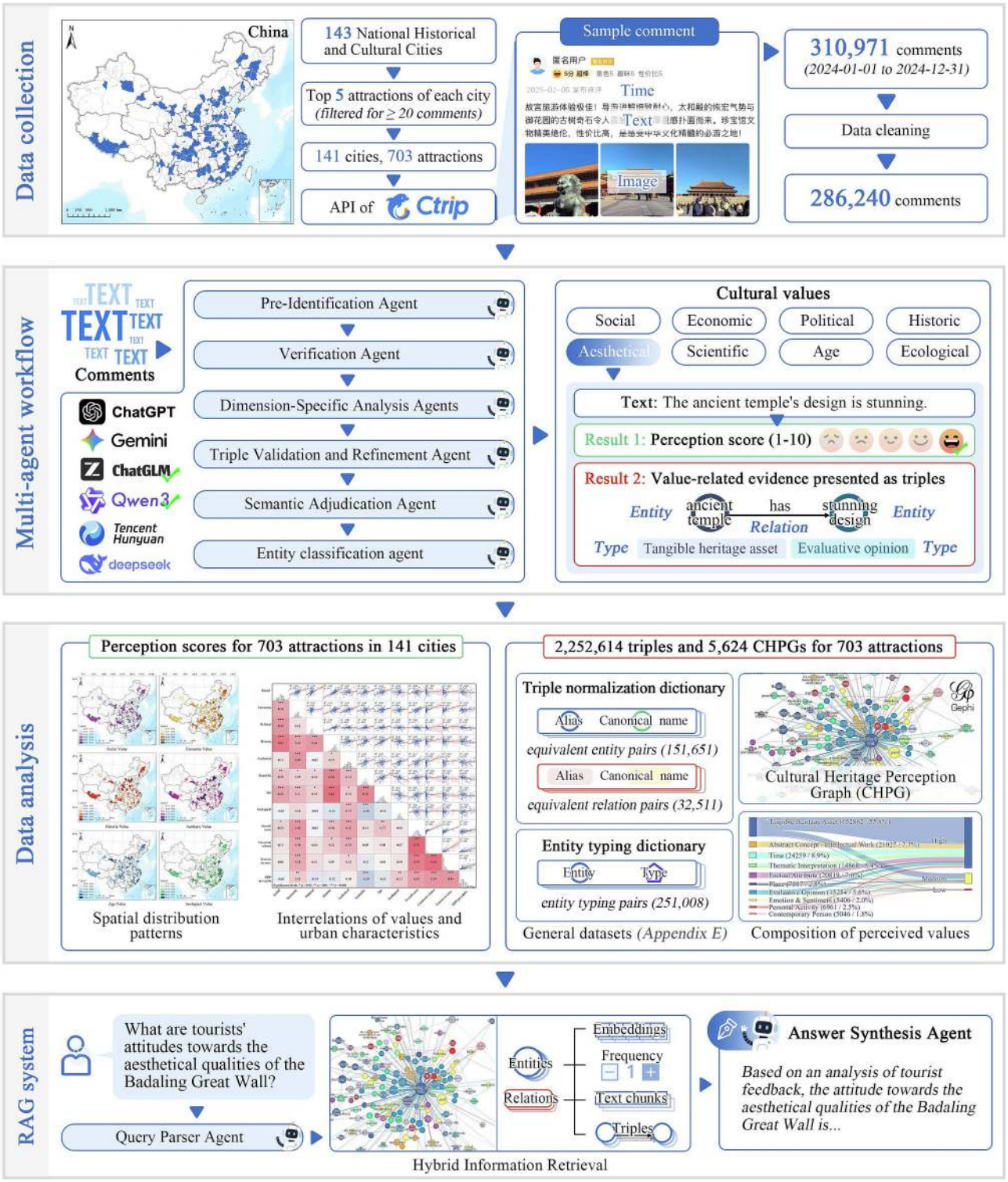


Fig. 2. Research workflow.

1) **Triple Validation and Refinement Agent:** The extracted triples may contain hallucinations, structural inaccuracies, or excessive complexity. This agent validates and refines the triples through several functions:

Contextual anchoring: Ensures each triple is explicitly linked to the primary heritage attraction it describes.

Coreference resolution: Resolves ambiguity by replacing pronouns and other vague references (e.g., "it", "here") with the precise entities they refer to (e.g., "the Great Wall").

Structural Normalization: Enforces a strict <Entity/Subject, Relation, Entity/Object> structure.

Atomic decomposition: Breaks down complex triples into simpler, more granular ones to ensure each triple represents a single piece of information.

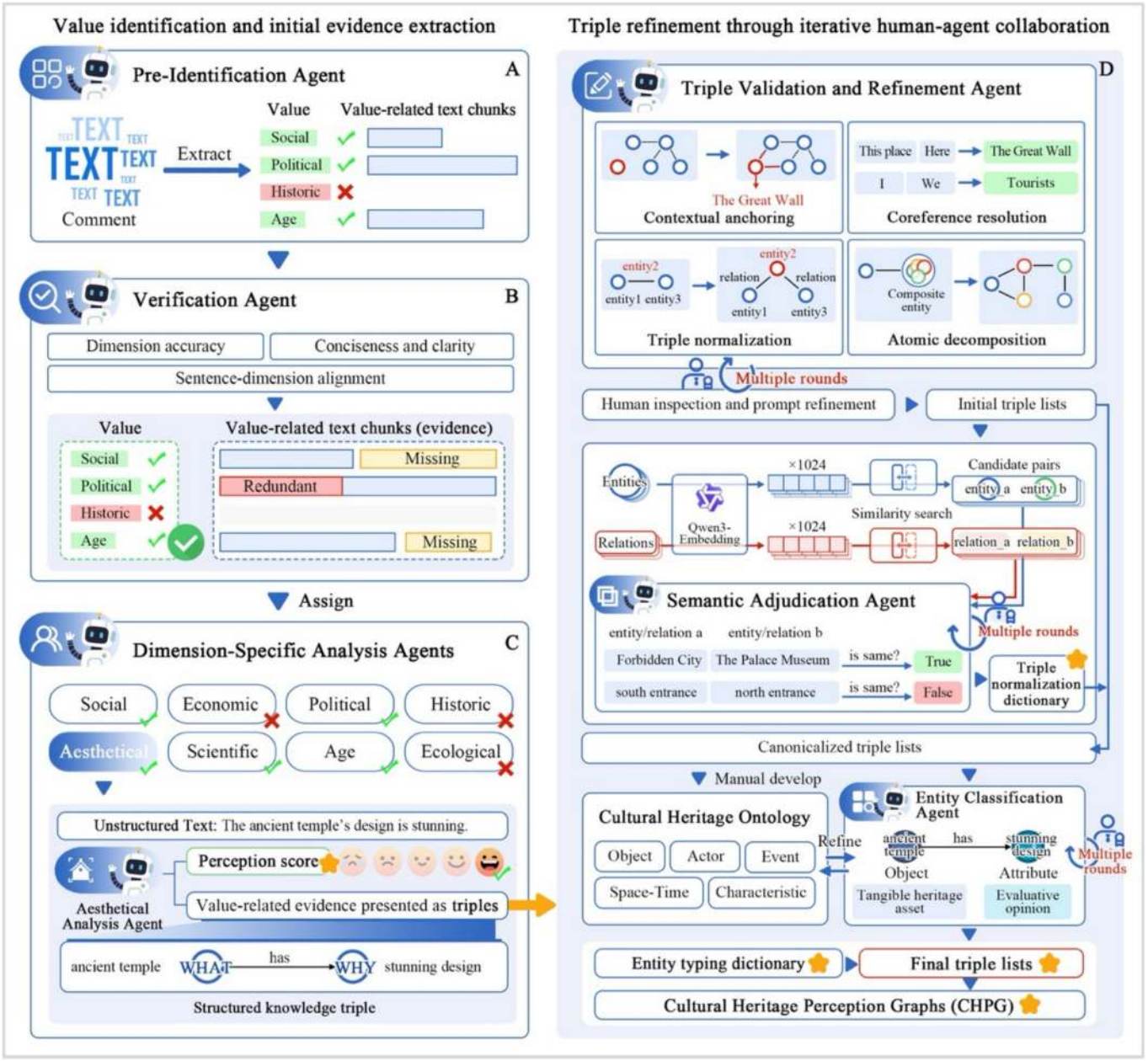


Fig. 3. Multi-agent LLM workflow for perceived cultural values scoring and value-related evidence extraction in triple form. Final outputs used in the analysis, including perception scores, the triple-normalization dictionary, the entity-typing dictionary, the final triple lists, and the CHPGs, are marked with stars (★). The left panel (A-C) presents the fully automated pipeline for value identification, perception scoring, and initial triple extraction. The right panel (D) presents triple postprocessing. Stages labeled "multiple rounds" denote iterative cycles of human review and targeted prompt refinement, through which the agents were progressively improved to extract and refine triples that satisfy the quality requirements of KG construction.

2) Entity and Relation Normalization.

To further standardize the extracted triples, all entities and relations undergo a normalization process:

Candidate pair identification: All unique entities and relations from the initial triple lists are converted into 1024-dimensional vectors using Qwen3-Embedding, then L2-normalized to unit length. An exact top-k nearest-neighbor search is performed using FAISS with an IndexFlatIP index. A pair (a, b) is considered a candidate if b is among a's top-k neighbors and their similarity score $s(a, b) \geq \theta$. This similarity is computed using cosine similarity as:

$$s(a,b) = \frac{\mathbf{v}_a^{\mathrm{T}} \mathbf{v}_b}{\|\mathbf{v}_a\|_2 \|\mathbf{v}_b\|_2} \tag{5}$$

Semantic adjudication agent: Candidate pairs are presented to this agent to determine if they refer to the same real-world entity or relation. If so, it designates a single canonical name. For example, "Forbidden City" and "Palace Museum" would be mapped to one identifier. This function is modeled as:

$$f_{adjudge}(e_a, e_b) \rightarrow \{\text{is_same} : \text{bool}, \text{canonical_name} : \text{str}\} \tag{6}$$

This process is repeated over multiple rounds with increasingly strict prompts to resolve ambiguities. The process culminates in

comprehensive triple normalization dictionaries that are used to replace all entities and relations with their canonical names, resulting in canonicalized triple lists.

Entity Classification Agent: This agent assigns each entity from the canonicalized triple lists to one or more classes within a custom CHPO (Table 1, Fig. 4), which was specifically designed for entities extracted from SMD of heritage attractions. Its class structure was informed by established ontological frameworks, including CIDOC-CRM (https://cidoc-crm.org/cidoc-crm/; Doerr (2003)) and the Schema.org vocabulary (https://schema.org/). For example, the six primary classes (Object, Actor, Event, Time, Place, and Characteristic) align with CIDOC-CRM classes such as E18 Physical Thing and E28 Conceptual Object (corresponding to our Object), E39 Actor (Actor), E5 Event (Event), E52 Time-Span (Time), E53 Place (Place), and E55 Type (related to our Characteristic). The CHPO and the Entity Classification Agent were refined in tandem based on manual examination of the entity typing dictionary (Fig. 3). In each iteration, we reviewed the agent's classification results, identified categories that were insufficiently comprehensive or semantically incoherent, and revised the ontology structure or the classification prompts accordingly. Detailed definitions of all CHPO classes are available in *Appendix C*.

Table 1
The hierarchical structure of the CHPO. The detailed definitions are available in *Appendix C*.

Class	Sub-Class	Type	Examples
Object	Heritage Asset	Tangible Heritage Asset	Corner Tower, terraced field
		Intangible Heritage Asset	Peking Opera, paper-cutting
		Movable Artifact	Bronze ware, calligraphy work
	Natural Feature		Pine tree, monkey
	Visitor Facility		Visitor center, restroom
	Commercial Service		Souvenir, guided tour
	Digital Content		Short video, photo
	Abstract Concept / Intellectual Work		Confucianism, ancient poetry
	Historical Object		Epang Palace, the Old Summer Palace
	Fictional Object		The Treasure Basin, Monkey King's golden cudgel
Actor	Person	Contemporary Person	Tourist, tour guide
		Historical Person	Confucius, Qin Shi Huang
	Natural Feature	Fictional Character	Monkey King, Mulan
	Organization		Volunteer association, Dunhuang Academy
Event	Personal Activity		Taking photos, queuing
	Organized Event		Temple fair, closure for renovation
	Emergency Event		Flood, fire
	Historical Event		Beijing Olympics, the Long March
	Fictional Event		The Legend of the White Snake
Time			Tang Dynasty, National Day holiday
Place			Xi'an, south gate
Characteristic	Objective Property	Factual Attribute	Made of bronze, 100 m long
		Operational Attribute	60 RMB ticket price, closed on Mondays
		Contextual Condition	Sunny weather, crowded
	Subjective Perception	Emotion & Sentiment	Amazing, disappointing
		Evaluative Opinion	Overly commercialized
		Thematic Interpretation	Patriotism, profound charm

Other Entity: Entities not falling into any of the above categories, occasionally mentioned in the context of heritage.

For a given entity e, the agent identifies its appropriate class or classes within the ontology O. This step results in an entity typing dictionary and transforms the canonicalized triple lists into semantically-grounded final triple lists. The function is:

$$f_{classify}(e) \rightarrow \{C_1, C_2, \ldots\} \subset O \tag{7}$$

3.4. CHPG-powered RAG system

3.4.1. RAG knowledge base construction

To enable an interactive and context-aware query system, a distinct knowledge base $KB_{A,v}$ is constructed using NetworkX for each unique combination of a heritage attraction A and a value v. Each knowledge base integrates two data structures:

1) CHPG ($G_{A,v}$): A graph storing extracted triples, where nodes represent unique entities and edges represent the relations between them from the final triple lists.
2) Dense vector indices: Using the Qwen3-Embedding model to generate 1024-dimensional vector representations of entities and relations. To optimize search, separate FAISS indices are created for individual entities ($I^{ent}_{A,v}$) and relations ($I^{rel}_{A,v}$). Every indexed item is linked to its frequency, value-related text chunks, and triples.

3.4.2. Architecture of the CHPG-powered RAG system

CHPG-powered RAG system allows stakeholders to ask plain-language queries and receive synthesized, evidence-based answers. User queries are processed through a structured, two-stage pipeline to generate contextually rich answers.

1) **Query Parsing Agent:** This agent analyzes user's query and breaks it down into a structured format that includes the target attraction A and value v, a cross-dimensional query flag, the desired sentiment, and a list of sentiment-related query terms.
2) Hybrid information retrieval:

   Entity & relation search: The user's query and extracted query terms are converted into 1024-dimensional vectors using Qwen3-Embedding. These vectors are used to perform a top-k similarity search against the entity ($I^{ent}_{A,v}$) and relation ($I^{rel}_{A,v}$) indices to identify semantically similar entities (E_{sim}) and relations (R_{sim}).
   Semantic graph traversal: The system then searches the CHPG ($G_{A,v}$) for triples containing these semantically similar entities and relations. The resulting triples, $Triple_{\text{sem}}$, are ranked by frequency, and their value-related text chunks are retrieved.
   Answer Synthesis Agent: This agent synthesizes all the gathered information into a coherent and comprehensive answer that directly addresses the user's query.

For queries that span multiple values, this RAG system can also summarize feedback across the various values to provide a more holistic overview.

3.5. Statistical analysis of perception scores and CHPG

3.5.1. Analysis of perception scores

Perception scores for each heritage attraction were adjusted using a Bayesian average to account for data sparsity, then normalized to a 0–100 scale. City-level cultural value scores were calculated by averaging the scores of all attractions, weighted by the relative discussion

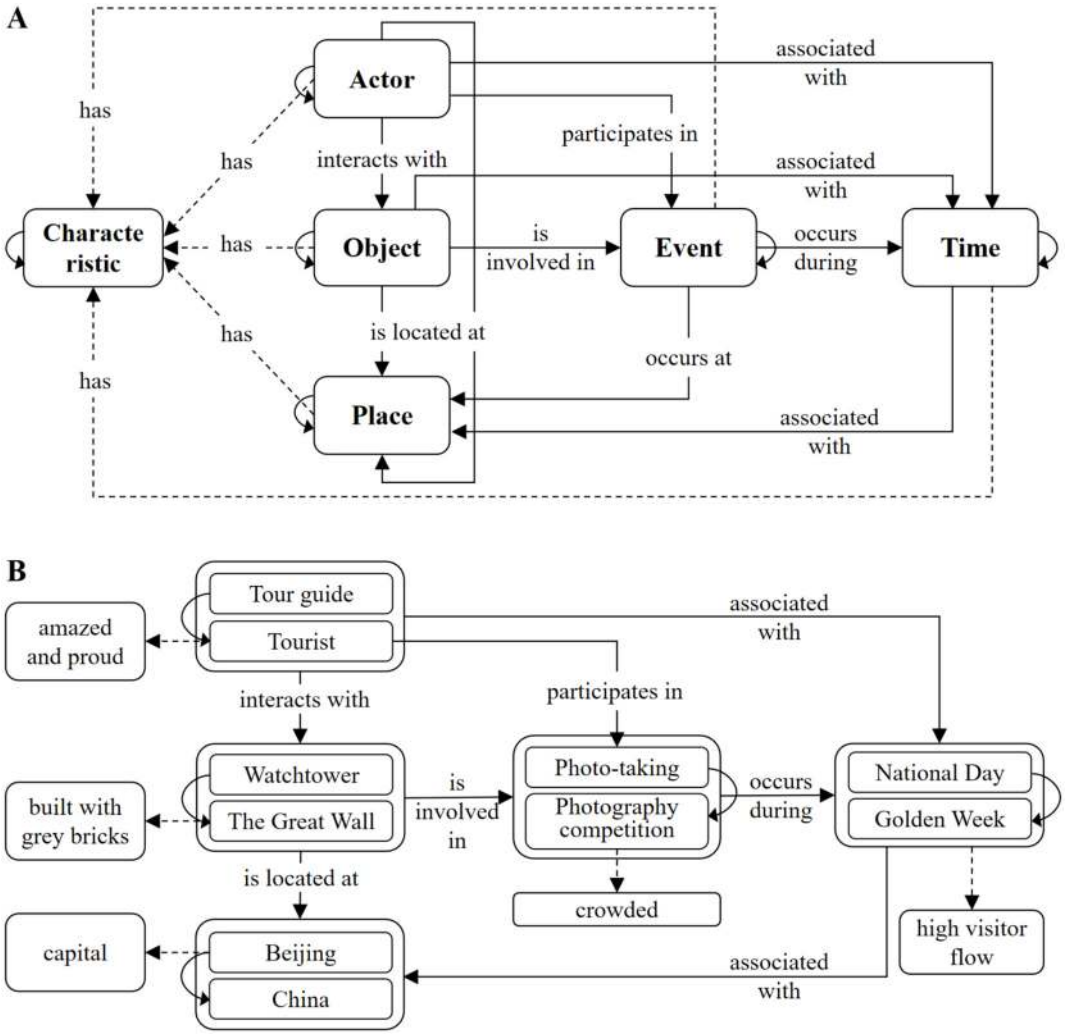


Fig. 4. Conceptual model and example of the CHPG. (A) The abstract model illustrates the relationships between the six core classes. (B) An example demonstrating how *ent*ities and relations are instantiated to represent tourist feedback about the Great Wall.

frequency of each value. These scores were combined with a city's online popularity (number of comments) to produce a final overall score. Detailed formulas are provided in *Appendix D*.

We used ArcGIS Pro to visualize the spatial distributions of perception scores across the 141 cities. To explore the urban characteristics associated with overall score, we compiled socioeconomic indicators for each city, including tourism revenue and GDP per capita in 2024. We then conducted a series of statistical analyses. First, Spearman correlation was performed to assess relationships among overall score, comment volume, tourism revenue, and GDP per capita. Next, a quadrant analysis based on median splits of overall score and GDP per capita classified cities into four typologies (e.g., high-score/high-GDP).

3.5.2. Analysis of CHPG

First, we analyzed the extracted triples from all 703 heritage attractions. For each value, we calculated the composition and sentiment distribution of entity types and visualized these patterns using Sankey diagrams. Next, recognizing that each attraction exhibits unique characteristics, we applied our multi-agent LLM workflow to generate 5624 attraction-specific CHPGs, creating one for each of the eight values per attraction. Taking the Badaling Great Wall as a case study, we visualized the overall CHPG structure with Gephi and examined relationships among entities within the network using Neo4j.

4. Results

4.1. LLMs evaluation results and model selection

The pricing and efficiency of the evaluated LLMs are detailed in

Table 2
LLM pricing and cost overview ($N = 1000$).

Model	Version date	Price (Output: / 1 M tokens)		Time (s)	Token
		CNY	USD		
GPT-4.1-mini	2025-04-14	11.39	1.60	18,451.31	3,195,590
Gemini-2.5-Flash	2025-04-17	17.80	2.50	43,000.83	8,132,781
DeepSeek-V3	2025-03-24	8.00	1.12	67,677.41	5,676,284
GLM-4-Plus	2024-08-29	5.00	0.70	39,703.97	6,136,016
GLM-4-Air	2025-04-14	0.50	0.07	17,513.47	5,822,216
Hunyuan-T1	2025-04-03	4.00	0.56	119,355.61	14,774,332
Qwen3-32b	2025-04-28	8.00	1.12	97,265.89	7,188,139
Qwen-Turbo	2025-04-28	0.60	0.08	13,003.74	4,752,429

Note: The processing times reported in this table were measured via direct API calls and are influenced by external factors such as network latency, server load, and API traffic. Therefore, these figures are intended for practical comparison and may not reflect the models' absolute inference speeds under ideal conditions. Additionally, using batch processing APIs could improve throughput and reduce costs.

Table 2. Among the tested models, Qwen-Turbo, GLM-4-Air, and GPT-4.1-mini demonstrated the fastest response times, with GPT-4.1-mini also offering the lowest token cost. In contrast, Hunyuan-T1 and Qwen3-32b were substantially slower, suggesting higher computational overhead and reduced practicality for large-scale deployment. Performance varied across perception scoring and value-related triple extraction (Table 3). For perception scoring, DeepSeek-V3 was the most accurate, achieving the lowest MAE (0.392) and MSE (0.714), followed closely by GLM-4-Plus, Hunyuan-T1, and Gemini-2.5-Flash. Qwen-Turbo and GPT-4.1-mini recorded higher error rates, indicating weaker performance in this task. In the value-related triple extraction, different models excelled in specific metrics: Gemini-2.5-Flash excelled in dimensional relevance (7.876), DeepSeek-V3 in information completeness (7.861), and GLM-4-Plus in content accuracy (8.317). DeepSeek-V3 ranked first or second in nearly all performance indicators, making it the most consistently high-performing model overall.

However, applying the models to the full dataset of 284,030 comments required a more practical balance of performance, speed, and cost. Although DeepSeek-V3 and Gemini-2.5-Flash achieved strong results, their slower speeds and higher costs made them less feasible for large-scale use. Ultimately, GLM-4-Air was selected as the operational model. It ranked second in speed and had the lowest token cost at just 0.50 CNY per million tokens. While not the overall leader, GLM-4-Air performed reliably across tasks and led in specific subdomains. This balance of efficiency, affordability, and competitive performance made it the most suitable choice for large-scale SMD analysis. Processing all comments with GLM-4-Air consumed approximately 1.24 billion tokens in total.

To verify that this choice does not compromise the reliability of our findings, we conducted a cross-model validation by processing four cities (20 heritage attractions, 7597 comments) using both GLM-4-Air and the best-performing DeepSeek-V3. At the attraction level, the two models yielded a Pearson correlation of 0.9872 ($p < 0.001$), with a mean score difference of only 0.38 points on the 0–100 scale. At the city level, the mean absolute difference was only 1.62 points (*Appendix G,* Fig. S1 and Table S3). These results confirm that the choice of model does not materially alter our substantive conclusions at the aggregated level.

4.2. Analysis results of perception scores

4.2.1. Statistical and spatial distribution of perception scores

The statistical and spatial distributions of perception scores for 141 historical cities are presented in Table 4 and Fig. 6. The most frequently mentioned dimensions are aesthetical and social values, with mention rates of 66.75% and 65.78%, respectively. Cities with high aesthetical value form several clusters: Eastern China (e.g., Shanghai, Suzhou, Yangzhou), Southwestern China (e.g., Dali, Leshan, Chongqing), and Northern China (e.g., Beijing, Pingyao, Datong). Cities with high social

Table 3
Model performance metrics across various domains (N = 1000).

Metric	Model	Social	Economic	Political	Historic	Aesthetical	Scientific	Age	Ecological	Average
Perception scoring										
MAE	Gemini-2.5-Flash	0.253	0.541	0.659	0.592	0.454	0.587	0.322	0.342	0.469
	GPT-4.1-mini	0.731	0.880	0.838	0.988	0.662	0.779	0.711	0.724	0.789
	DeepSeek-V3	0.310	**0.308**	**0.421**	**0.511**	**0.418**	**0.297**	0.458	0.416	**0.392**
	GLM-4-Plus	0.309	0.515	0.483	0.529	0.465	0.327	**0.272**	**0.296**	0.400
	GLM-4-Air	0.633	0.750	0.769	0.589	0.462	0.721	0.708	0.777	0.676
	Hunyuan-T1	**0.213**	0.547	0.667	0.592	0.440	0.585	0.295	0.348	0.461
	Qwen3-32b	0.710	0.874	0.814	0.975	0.647	0.769	0.698	0.689	0.772
	Qwen-Turbo	1.279	0.679	0.904	0.834	0.600	1.420	0.298	1.852	0.983
MSE	Gemini-2.5-Flash	0.649	1.011	0.948	0.841	0.551	0.912	0.475	0.548	0.742
	GPT-4.1-mini	1.651	1.891	1.651	2.260	1.103	1.645	2.055	1.791	1.756
	DeepSeek-V3	0.870	**0.750**	**0.520**	**0.769**	0.671	**0.401**	0.921	0.812	**0.714**
	GLM-4-Plus	0.659	1.095	0.748	0.805	0.724	0.723	0.551	**0.470**	0.722
	GLM-4-Air	0.804	0.926	0.911	0.872	0.667	0.954	1.241	1.278	0.957
	Hunyuan-T1	**0.610**	1.005	0.963	0.838	**0.541**	0.906	**0.462**	0.483	0.726
	Qwen3-32b	1.593	1.856	1.612	2.182	1.017	1.620	1.993	1.756	1.704
	Qwen-Turbo	3.620	1.639	2.288	0.908	0.832	4.715	0.638	4.982	2.453
Value-related triple (value evidence) extraction										
Dimensional relevance	Gemini-2.5-Flash	8.291	8.055	6.511	8.151	8.194	8.123	**7.703**	7.981	**7.876**
	GPT-4.1-mini	6.006	6.845	6.657	7.505	5.701	7.499	5.642	7.329	6.648
	DeepSeek-V3	7.551	8.492	6.673	**8.319**	9.088	7.291	7.484	8.018	7.865
	GLM-4-Plus	**8.808**	**8.994**	6.562	7.829	8.209	7.136	6.892	7.710	7.768
	GLM-4-Air	8.608	7.592	**6.800**	7.101	**9.215**	6.819	6.389	7.893	7.552
	Hunyuan-T1	7.265	8.072	6.484	8.193	8.010	**8.349**	7.455	**8.024**	7.732
	Qwen3-32b	6.039	7.976	6.669	7.656	6.763	7.582	6.658	7.237	7.073
	Qwen-Turbo	4.655	6.490	6.041	5.939	6.320	5.885	5.800	7.277	6.051
Information completeness	Gemini-2.5-Flash	8.011	8.289	8.378	8.422	7.401	7.672	**7.230**	5.907	7.664
	GPT-4.1-mini	6.223	7.499	8.510	7.979	7.311	7.174	5.623	5.178	6.937
	DeepSeek-V3	**8.433**	**9.522**	8.549	**8.439**	8.639	6.933	6.710	5.663	**7.861**
	GLM-4-Plus	6.920	8.117	8.593	8.070	**8.932**	7.021	7.018	5.841	7.564
	GLM-4-Air	8.234	8.531	6.743	7.242	8.705	6.560	5.714	**6.004**	7.217
	Hunyuan-T1	6.725	8.401	8.469	8.406	7.251	**7.940**	7.094	5.897	7.523
	Qwen3-32b	6.033	7.891	**9.599**	8.081	6.383	7.260	6.611	5.237	7.137
	Qwen-Turbo	4.487	6.561	7.889	6.218	5.723	6.588	5.576	5.258	6.038
Content accuracy	Gemini-2.5-Flash	8.401	8.243	8.557	9.229	7.274	9.296	7.541	6.922	8.183
	GPT-4.1-mini	6.002	7.968	8.601	8.677	5.714	8.588	5.680	6.211	7.180
	DeepSeek-V3	**8.907**	**9.112**	8.575	8.952	8.300	8.155	6.932	6.719	8.207
	GLM-4-Plus	7.650	8.532	**8.689**	**9.402**	9.305	8.349	**7.571**	7.038	**8.317**
	GLM-4-Air	8.831	8.703	6.627	8.229	**9.399**	7.803	6.405	6.901	7.862
	Hunyuan-T1	7.392	8.262	8.497	9.370	7.144	**9.441**	7.564	**7.044**	8.089
	Qwen3-32b	6.041	8.584	8.688	8.723	6.796	8.629	6.666	6.237	7.546
	Qwen-Turbo	4.682	6.530	7.087	6.989	5.380	7.893	5.798	6.279	6.330

Note: The best-performing model is shown in bold, and the second-best is underlined.

Table 4
Statistics of perception scores.

Dimension	Mention count	Score 1–2 (%)	Score 3–4 (%)	Score 5–6 (%)	Score 7–8 (%)	Score 9–10 (%)	Average score
Social	189,752 (65.78%)	2.07	5.04	10.89	58.35	23.64	6.06
Economic	143,521 (49.75%)	3.30	12.95	32.50	47.20	4.04	6.01
Political	12,645 (4.38%)	6.30	16.29	20.13	56.88	0.40	6.80
Historic	87,579 (30.36%)	1.25	3.93	12.43	68.62	13.77	7.49
Aesthetical	192,549 (66.75%)	2.80	3.21	25.43	55.11	13.45	7.55
Scientific	47,262 (16.38%)	11.61	18.40	39.36	28.09	2.53	5.06
Age	49,137 (17.03%)	2.89	10.33	41.56	41.75	3.47	7.18
Ecological	18,986 (6.58%)	0.95	3.75	21.80	70.45	3.05	6.00

value scores typically feature heritage that embodies significant spiritual beliefs, collective memories, and cultural identity. Top-ranked examples include Qufu (Temple and Cemetery of Confucius and the Kong Family Mansion), Beijing (the Palace Museum), and Qi County (Qiao Family Compound).

Economic value, mentioned in 49.75% of cases, is highest in major, highly developed cities like Beijing and Shanghai, followed by other key cities in Eastern China like Hangzhou, Nanjing, and Suzhou, as well as other metropolises such as Chongqing, Wuhan, and Guangzhou. Although political and ecological values were mentioned far less frequently (4.38% and 6.58%, respectively), they exhibit the clearest spatial concentrations. High political value is concentrated in cities significant to China's revolutionary history (e.g., Beijing, Ruijin, Yan'an, Zunyi) or those with attractions symbolizing political authority (e.g., Shenyang Imperial Palace). High ecological value is primarily located in the coastal southeast and mountainous southwest, where heritage attractions are recognized for their perceived harmony with the natural environment.

Historic and age values share similar spatial patterns, with the highest scores in ancient capitals with monumental attractions like Pingyao (Zhenguo Temple), Datong (Yungang Grottoes), Dunhuang (Mogao Caves), Xi'an (Terracotta Army), and Luoyang (Longmen Grottoes). Scientific value scores are highest at specific attractions celebrated for exceptional "workmanship" and "technological skillfulness," such as Datong (Hanging Temple), Pingyao (Shuanglin Temple), Dunhuang (Mogao Caves), Zhengding (Longxing Temple), and Suzhou (Lion Grove Garden).

4.2.2. Relationships between perception scores and urban characteristics

As shown in Fig. 7, ecological value stands out as a distinct dimension. While it is moderately positively correlated with aesthetical value ($r = 0.33$) and weakly correlated with economic value ($r = 0.16$), it exhibits slight negative correlations with most other values. Notably, it is negatively associated with scientific value ($r = -0.30$). In contrast, the other seven values are generally positively correlated with one another, indicating a high degree of consistency in how different values are co-perceived. Comment volume is strongly correlated with both tourism revenue ($r = 0.68$) and GDP per capita ($r = 0.49$). Economic and aesthetical values are positively associated with both comment volume and tourism revenue, whereas other values show no similarly strong relationships. Interestingly, while economic and ecological values are positively correlated with GDP per capita, most other values, particularly scientific value, are negatively correlated with it ($r = -0.29$).

The scatter plot of overall score and GDP per capita in Fig. 7 is enlarged in Fig. 8, where a quadrant analysis classifies cities into four typologies. In the High-Score/High-GDP quadrant (top right), cities such as Beijing, Shanghai, Suzhou, Nanjing, and Hangzhou rank among the highest in both perception scores and GDP. These cities host prominent heritage attractions, including the Palace Museum in Beijing and the exotic architectural clusters along Shanghai's Bund, both of which score high across multiple values and attract the highest volumes of tourist comments. The High-Score/Low-GDP quadrant (top left) includes cities such as Luoyang, Dali, Datong, and Pingyao, which achieve high overall scores despite below-median levels of economic development. In contrast, cities such as Yulin, Changzhou, Yixing, and Taizhou fall into the Low-Score/High-GDP quadrant (bottom right), showing relatively low scores despite higher levels of GDP. Finally, the Low-Score/Low-GDP quadrant (bottom left) includes cities that score below the median in both heritage perception and economic indicators.

4.3. Results of value-related triple extraction and CHPG construction

After multiple rounds of processing with the Triple Validation and Refinement Agent, Semantic Adjudication Agent, and manual validation, the number of entities was reduced from 458,030 to 251,056, and relations from 119,873 to 50,817. Specifically, we only merged identical terms, such as "the Palace Museum" and "the Forbidden City". Similar terms with high cosine similarity but different meanings were preserved to maintain the diversity of user-generated expressions. These processes produced two triple normalization dictionaries containing 151,651 entity pairs and 32,511 relation pairs (available in *Appendix E*). Subsequently, the Entity Classification Agent categorized each of the 251,008 entities according to our CHPO, resulting in an entity typing dictionary (available in *Appendix E*).

4.3.1. All triples: Frequently mentioned aspects of each dimension

Analysis of all triples (value evidence) (Fig. 9) reveals that tangible heritage assets (e.g, building and architectural elements) are the central focus of public discourse, aligning with the findings of Foroughi et al. (2023) and Ginzarly and Srour (2022). However, perceptions tend to concentrate on specific aspects within each value. For social value, the most frequently mentioned aspects involve contemporary persons (16.1%), such as tour guides, children, and elderly visitors, followed by descriptions of their activities (10.0%), including visiting, understanding, experiencing, and photo-taking. Evaluative opinions (11.8%) also feature prominently, using terms like "worth-visiting," "unique," "beautiful," and "rich history." Comments on economic value predominantly concern the direct use value of heritage, often measured through market transactions. Commercial services (10.4%), including tickets, restaurants, cultural and creative products, and parking fees are frequently discussed, along with evaluative opinions (15.6%) on pricing, such as "too expensive," "cheap," "worthy," or "reasonable." Attractions rated highly for economic value are often described as offering tickets "worthy" of the experience.

For political value, comments center on abstract concepts (12.8%), such as "Chinese history," "national symbol," "sovereignty," and "Chinese nation," along with factual attributes (9.1%) denoting official status, including "National Model Site for Patriotism Education," "sacred place for the revolution," and "iconic landmark." Historic value is conveyed through abstract concepts (11.1%), such as "history," "culture," "story," and "historic knowledge," supported by factual attributes (7.1%) that confirm its significance, such as "World Cultural Heritage" designation, "long history," "witness of history," and "historical background."

For aesthetical value, evaluative opinions dominate (15.5%), often describing an attraction as "beautiful," "grand sight," "pleasant," or "great." Discussions of scientific value focus on factual attributes (7.9%) like "height," "characteristic," "design," "material," and "structure,"

alongside evaluative opinions (8.9%) highlighting craftsmanship and scale, using terms like "exquisite," "excellent craftsmanship," "giant," "true to life," and "magnificent."

Perceptions of age value revolve around expressions of time (8.9%), referencing "over millennia," "ancient China," "Ming and Qing Dynasty," and other dynasties such as Tang, Song, Sui, Yuan, and Han. For ecological value, natural features (14.9%), such as stones, trees, rivers, and lakes, are most frequently mentioned, followed by opinions (8.7%) on natural beauty and contextual conditions (6.7%), such as weather, season, and other visit-related factors.

4.3.2. *Attraction-level insights from CHPGs*

The most valuable application of the value-related triples is the construction of CHPGs for each of the eight values at all 703 attractions, resulting in a total of 5624 CHPGs. These provide detailed, attraction-level insights into public perception. Using the Badaling Great Wall as an example, a CHPG (Fig. 10) is a network graph centered on the heritage asset ("Badaling Great Wall"), surrounded by nodes representing related entities. The connecting lines indicate relations between entities, with attributes such as name and frequency (Fig. 11). A higher frequency for a node or relation indicates it was mentioned more often in the SMD, representing a more widely held viewpoint. This structure enables fine-grained analysis, such as identifying specific negative feedback where "scaffolding wrap" is shown to affect "taking photos" (Fig. 12).

As shown in Fig. 12, CHPG complexity varies significantly across different values. The social value graph, for instance, is dense and centered on interactions between people (e.g., "tourist," "tour guide") and the attraction. In contrast, other values have distinct central entities and relationships. This demonstrates the multi-agent LLM workflow's ability to differentiate values and extract relevant, structured evidence from SMD.

These CHPGs are not static visualizations but form the dynamic knowledge base for our CHPG-powered RAG system (Fig. 5). When a user asks a question such as *"What are tourists' attitudes towards the aesthetical qualities of the Badaling Great Wall?"*, the Query Parser Agent identifies key information such as the attraction name, value dimension, and query terms. This information guides a hybrid information retrieval process that fetches high-frequency value-related triples and corresponding text chunks from the knowledge base. The Answer Synthesis Agent then uses this contextualized information to generate a comprehensive, evidence-based response.

Beyond single-value interpretation, the CHPG-powered RAG system supports a variety of complex information needs that reflect real-world decision-making scenarios. For instance, it can address diagnostic queries that identify specific factors diminishing visitor experience, such as *"What factors negatively influence the social experience at the Chengdu Wuhou Shrine?"* The system retrieves evidence linking negative sentiments to entities such as "crowding," "long queuing," and "insufficient rest areas," enabling managers to pinpoint actionable issues. Second, it supports explanatory queries across different values. For example, asking *"Why do tourists perceive strong historic value at the Yungang Grottoes?"* retrieves triples describing dynastic origins, significant historical events, and iconic carvings that shape visitors' interpretations. Third, it can synthesize multi-value insights, as in *"How do ecological and aesthetical perceptions interact at West Lake?"*, returning evidence about seasonal scenery, lakeside vegetation, and environmental conditions that jointly influence visitor sentiment. Finally, the system enables comparative analysis across attractions, such as *"How do the economic perceptions of the Terracotta Army differ from those of the Longmen Grottoes?"*, summarizing differences in ticket pricing, perceived value

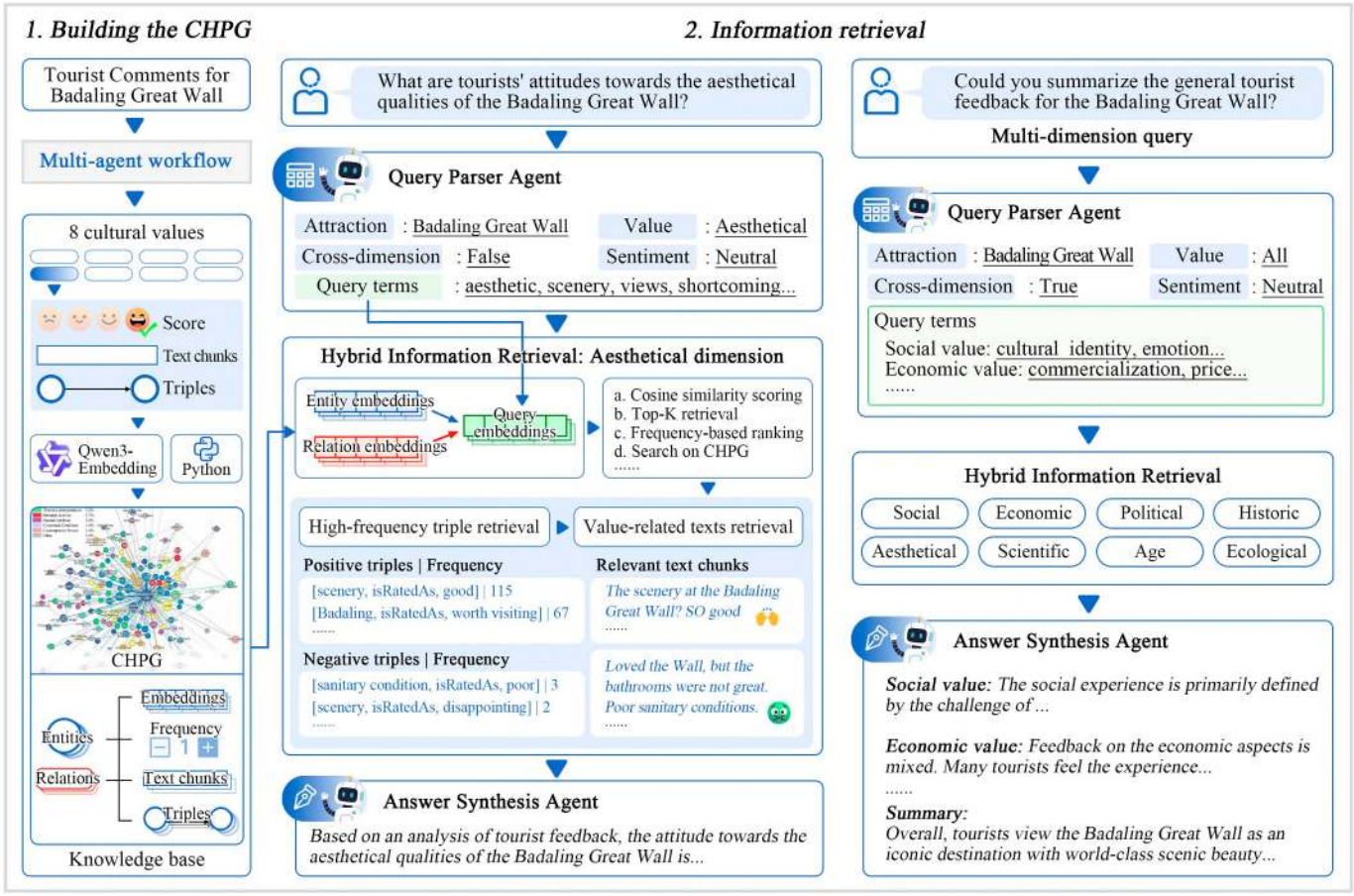


Fig. 5. Architecture of the CHPG-powered RAG system.

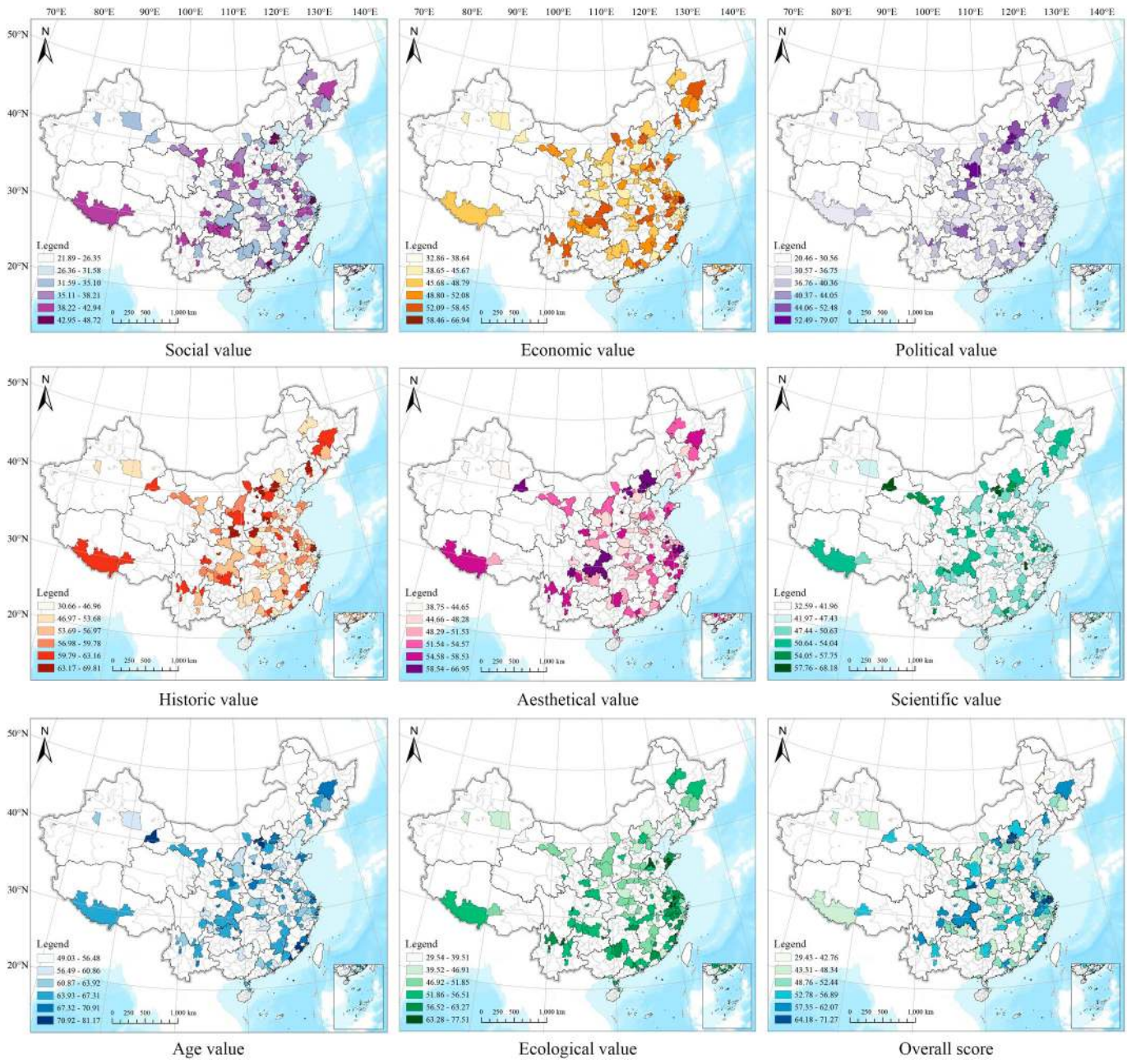


Fig. 6. Spatial distributions of perception scores among the 141 historical cities.

for money, and commercial services.

5. Discussion

Understanding the cultural values perceived by the public is fundamental to sustainable heritage management (Jones, 2017; UNESCO, 2011). SMD has become a vital resource for this, capturing how people share memories, thoughts, and feelings that keep heritage alive in the collective consciousness (Ginzarly & Srour, 2022). Analyzing these public expressions against official discourses is crucial for identifying gaps between perceived and expert-defined values, thereby informing more responsive heritage management (Bai et al., 2023; Bai, Nourian, & Roders, 2021; UNESCO, 2011).

5.1. Methodological advances and observations: From a single LLM to a collaborative multi-agent LLM workflow

Our research advances the analysis of public-perceived cultural values by moving from single-model approaches to a collaborative multi-agent LLM workflow. First, instead of generating generic sentiment scores from SMD, our workflow identifies eight cultural values following a well-established typology (Roders, 2007; Silva & Roders, 2012). It effectively identifies perceived cultural values and extracts, structures and operationalizes the evidence behind them through NLP tasks such as text classification, information extraction, sentiment analysis, named entity recognition, and relation extraction. Crucially, this LLM-based workflow performs these functions without the labor-intensive data annotation, complex model training, or high computational costs typical of supervised learning models (Bai et al., 2023; Fan & Chen, 2024; Fan & Wang, 2022; Koufodontis & Gaki, 2022; Wang, Liu, et al., 2024; Zhao et al., 2024).

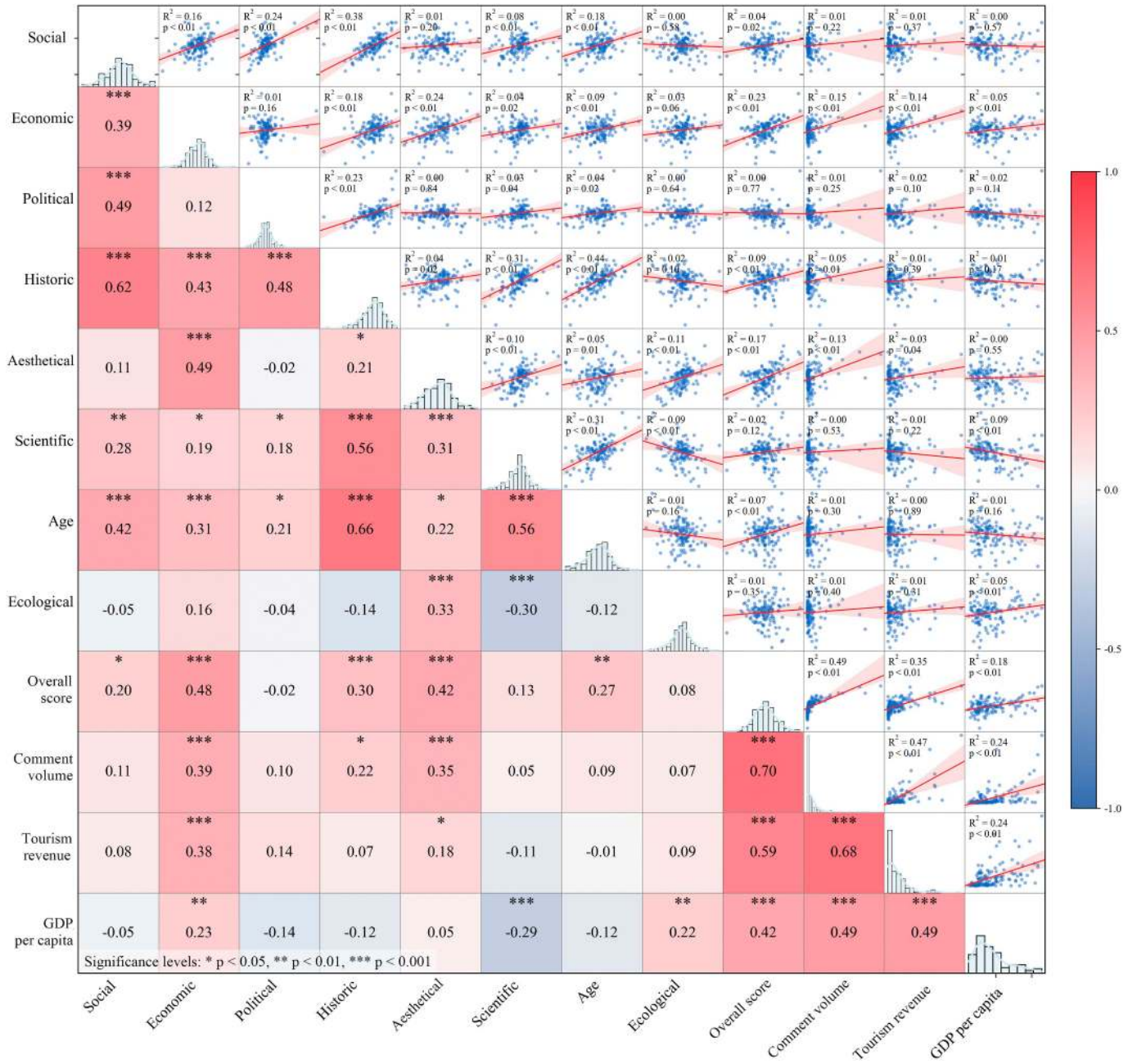


Fig. 7. Correlation matrix of perception scores, comment volume, tourism revenue, and GDP per capita. The lower triangle shows Spearman's rank correlation coefficients with significance levels. The diagonal displays histograms for each variable's distribution. The upper triangle contains scatter plots with linear regression fits, showing the coefficient of determination (R^2) and *p*-value.

Second, we move beyond text analysis methods such as word frequency analysis, topic modeling, and clustering. While these methods effectively identify key factors and themes (Boota et al., 2025; Ginzarly & Srour, 2022; Ning & Chen, 2023; Zhao et al., 2024), their outputs often remain lists of isolated terms that lack contextual relationships. To address this, we construct structured, interactive CHPGs that map value-related entities and their relations for each value at every attraction. These CHPGs power a RAG system that allows stakeholders to ask plain-language questions and receive synthesized, evidence-based answers. This approach bridges the gap between raw data and actionable insights and aligns expert and public understandings of heritage (Bai et al., 2023; Ginzarly et al., 2019). Looking forward, integrating CHPGs with KGs built on expert-curated knowledge (Bai et al., 2022; Carriero et al., 2019; Fan & Chen, 2024; Fan & Wang, 2022; Pellegrino et al., 2022; Wang, Liu, et al., 2024) could enable more holistic, evidence-based decision-making and enhance personalized tourism recommendations (Fan & Chen, 2024; Wang, Liu, et al., 2024). Leveraging the human-like perceptual capabilities of LLMs, our workflow has the potential to be extended to other types of built environments and adapted for data from different social media platforms, countries, regions, and languages (Bai et al., 2023; Boota et al., 2025; Luo et al., 2025).

Furthermore, we note two key observations: 1) Our attempts to use highly advanced LLMs (e.g., Gemini-2.5-Pro, OpenAI o3) in early 2025 for all tasks resulted in hallucinations, misjudgments, and outputs requiring manual correction. Decomposing the process into constrained, operationally defined sub-tasks enables agents to repeatedly verify and build upon the outputs of preceding agents, allowing each to focus on simpler, well-defined goals. This achieves effective performance even

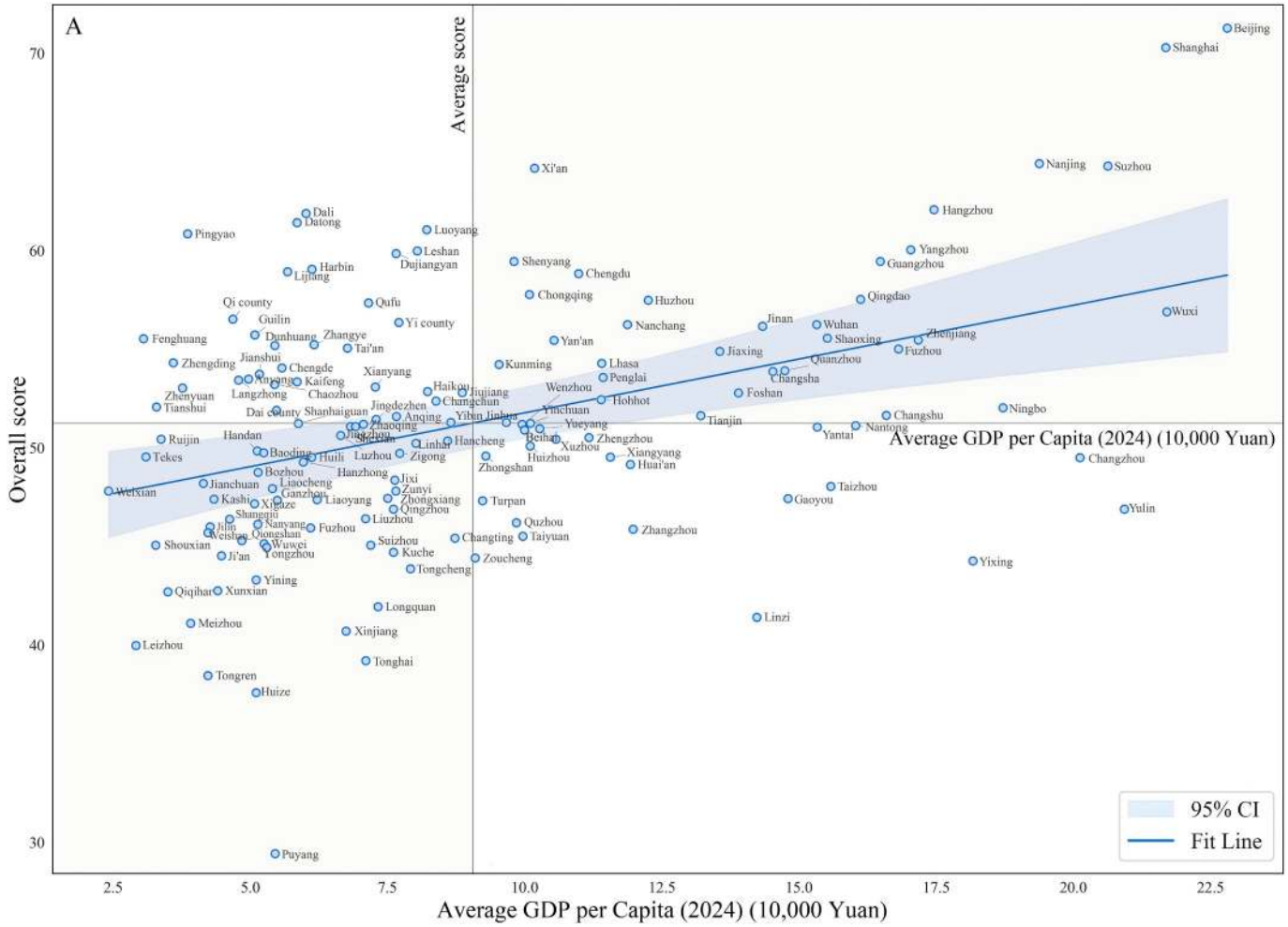


Fig. 8. Relationships between overall score and GDP per capita.

with cost-effective models like GLM-4-Air (Table S2). Although more advanced but considerably more expensive LLMs available in 2026 (e.g., GPT-5.4 Pro, Gemini 3.1 Pro, Claude 4.6 Opus) may achieve comparable or stronger performance in a single pass with well-designed prompts, our study demonstrates the possibility of using smaller, faster, and more affordable models to achieve satisfactory results through multi-agent task decomposition. 2) While some multi-agent systems use LLMs as supervisors (Chen et al., 2023; Lu et al., 2026), current LLMs struggle to ensure quality and consistency across hundreds of thousands of triples. Manual validation is still necessary to refine and normalize outputs, ensuring the integrity of the knowledge base.

5.2. Empirical insights

Across 141 cities, aesthetical (66.75%) and social (65.78%) values were the most prominently perceived, forming clear spatial clusters in Eastern, Southwestern, and Northern China. Online discussions center on personal activities (e.g., visiting, photo-taking), subjective opinions (e.g., beautiful, worth-visiting) and social interactions with fellow visitors. This highlights the strength of SMD in capturing the personal connections people form with heritage, perspectives often absent from expert-driven assessments (Ginzarly et al., 2019; Jones, 2017). Such experiences are known to enhance satisfaction (Roders, 2007), fostering memorable connections and place attachment that encourage revisitation (Peng et al., 2023). To strengthen these dominant values, managers can enhance scenic viewpoints, manage crowd flow, and improve family-friendly and interactive services that facilitate positive onsite experiences.

Economic value was also prominent (49.75%), particularly in prosperous cities like Beijing and Shanghai. Supporting Foroughi et al. (2023)'s observation that public commentary emphasizes certain aspects of values, our analysis showed that discussions of economic value centered almost exclusively on direct-use and entertainment aspects such as ticket pricing and perceived cost-benefit, while non-use value was rarely mentioned. Economic score evaluated from SMD complements conventional economic valuation methods, which typically rely on willingness-to-pay or cost-based estimates, by adding a qualitative dimension that captures real-time public sentiment.

Historic and age values showed a strong correlation ($r = 0.66$) and scored highly in ancient capitals like Pingyao, Datong, and Dunhuang. Historic value emphasized intangible narratives such as important events, stories, or people, whereas age value related to temporal depth and dynastic chronology (Roders, 2007; Silva & Roders, 2012). Scientific value appeared less frequently, typically associated with exceptional craftsmanship or unique architectural features. It reached high scores at only a few heritage attractions (e.g., the Hanging Temple, Shuanglin Temple). Strengthening narrative interpretation and explaining distinctive construction techniques may enhance public appreciation of historic, age, and scientific values.

Political (4.38%) and ecological (6.58%) values were the least frequently mentioned but exhibited distinct spatial concentrations. Despite its low frequency, the political value was highly significant where it appeared. Discussions focused on abstract concepts like national symbols or official designations (e.g., Patriotism Education Site),

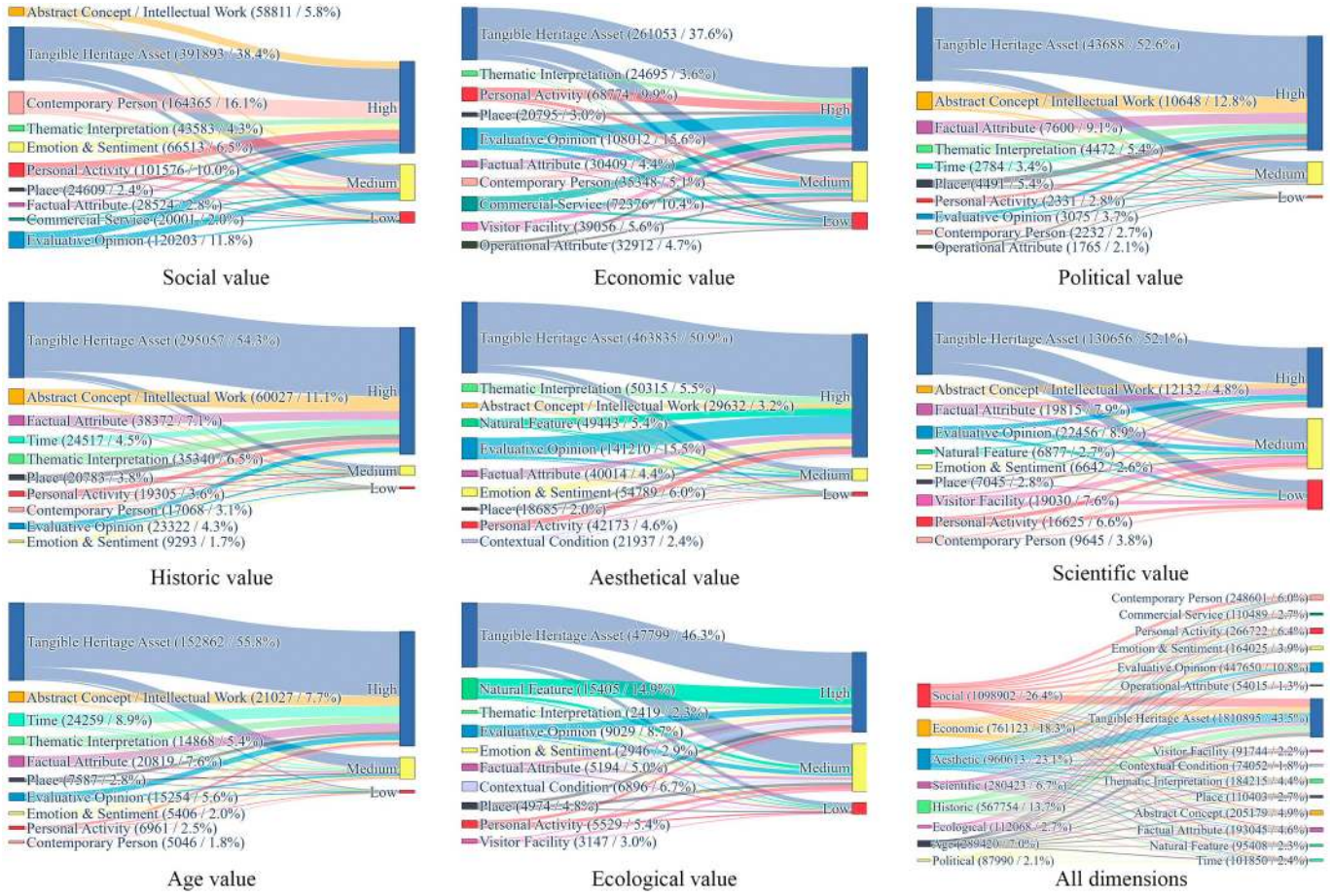


Fig. 9. Composition of perceived cultural values.

leading to high scores within its tightly clustered distribution in key cities of China's political history like Beijing, Yan'an, and Zunyi. Ecological value was also geographically concentrated in coastal and mountainous regions, with discussions emphasizing the perceived harmony between attractions and their natural surroundings (Roders, 2007; Silva & Roders, 2012). Interestingly, ecological value showed a significant negative correlation with scientific value ($r = -0.30$), suggesting a perceived tension between natural harmony and human ingenuity.

At the macro level, overall score positively correlates with comment volume, tourism revenue, and GDP per capita, indicating that economically prosperous cities tend to receive more positive online discussion. Cities in the High-Score/High-GDP quadrant, such as Beijing, Shanghai, and Suzhou, combine renowned heritage with robust tourism infrastructure, high media visibility, and sustained conservation investment, creating a self-reinforcing cycle of economic strength and cultural prominence. High-Score/Low-GDP cities like Luoyang, Datong, and Pingyao benefit from dense clusters of historic attractions and distinctive narratives. More strategic promotion, cultural branding, and infrastructure improvements could translate these cultural assets into stronger economic returns. Low-Score/High-GDP cities underutilize their financial resources for heritage marketing, storytelling, and visitor engagement, representing missed opportunities to elevate public appreciation. Finally, Low-Score/Low-GDP cities face the dual challenge of limited resources and low cultural recognition. Targeted revitalization efforts integrating conservation, tourism, and community participation could ignite cultural interest while supporting economic growth. Practical interventions include improving the destination's image, upgrading tourism infrastructure, and enriching educational content to enhance visitor satisfaction and appreciation (Havinga et al., 2020; Luo et al., 2025; Peng et al., 2023). Longer-term monitoring of public-perceived cultural values through our workflow could further improve heritage management strategies (Foroughi et al., 2023).

5.3. Limitations and future directions

First, while our workflow provides valuable insights from SMD, practitioners must exercise critical judgment when applying these findings to heritage management decisions. SMD-derived insights should complement, not replace, traditional assessment methods and expert knowledge. Second, our analysis provides only a temporal snapshot of public perception during 2024. While valuable, this does not capture the dynamic nature of heritage, which is a social construct that evolves over time (Foroughi et al., 2023; Ginzarly & Srour, 2022; Silva & Roders, 2012). Considering that SMD is generated in real-time (Boota et al., 2025; Ginzarly et al., 2019), future work could conduct longitudinal analysis to monitor how cultural values and visitation patterns change, particularly in response to significant events (Bai, 2023). This would provide authorities with more immediate feedback on the effects of their actions (Foroughi et al., 2023). Third, a key limitation is the inherent demographic bias of SMD. Ctrip users predominantly represent younger, digitally active tourists rather than the general public. Certain stakeholder groups, such as elderly visitors, local residents, business owners, and heritage professionals, are likely underrepresented or absent from this data source (Boota et al., 2025; Zhao et al., 2024). This demographic skew may influence the observed value patterns. For instance, the prominence of aesthetical and social values may partly reflect the experiential priorities of leisure tourists, who tend to emphasize visual appeal and social sharing. To foster more inclusive management, insights from our analysis should be integrated with qualitative data from traditional methods like on-site questionnaires and

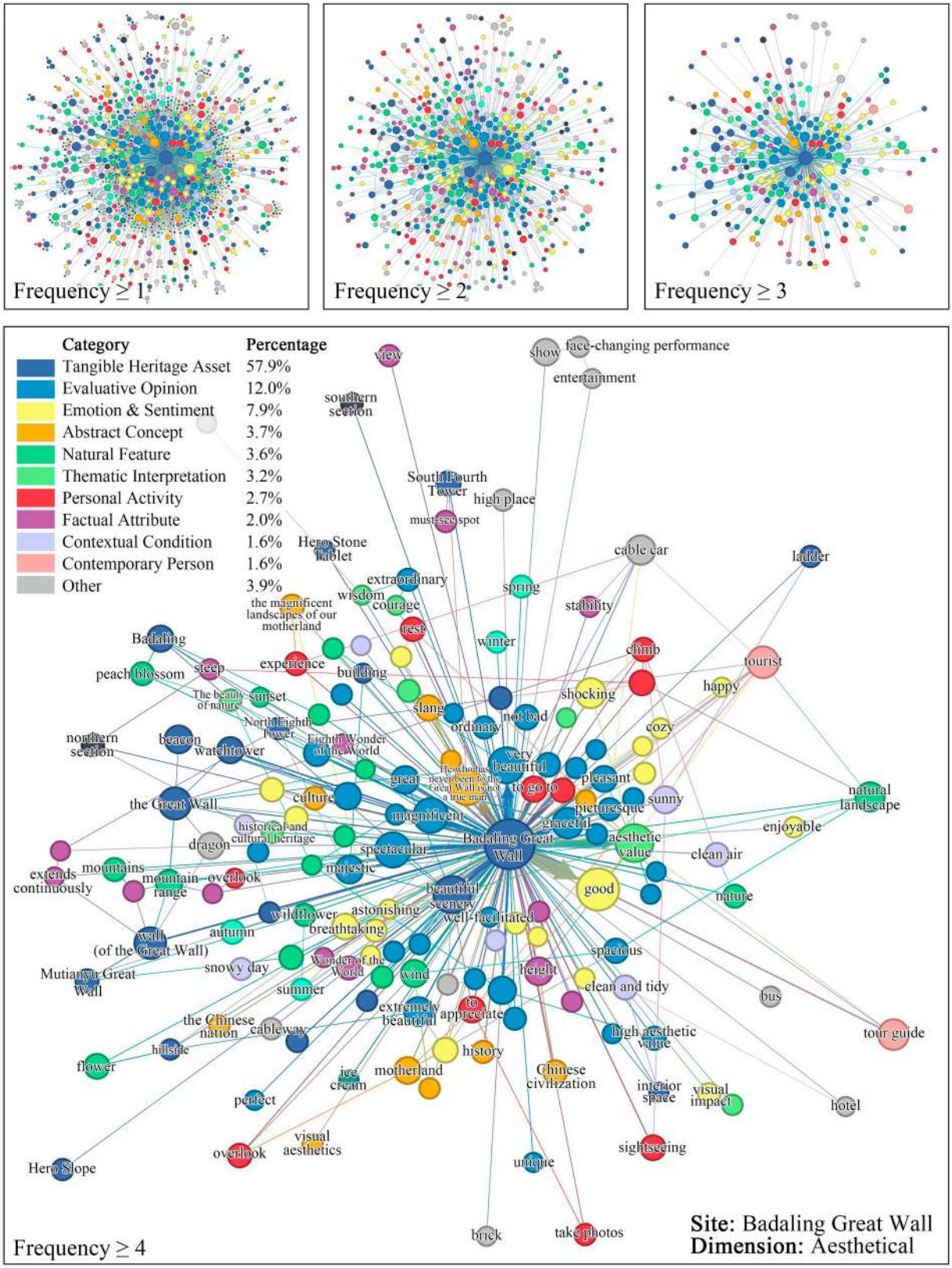


Fig. 10. CHPGs for the aesthetical value of the Badaling Great Wall. Nodes represent entities extracted from SMD, with larger nodes indicating higher frequencies, and are categorized by type. Edges represent relations between entities, with edge thickness proportional to relation frequency.

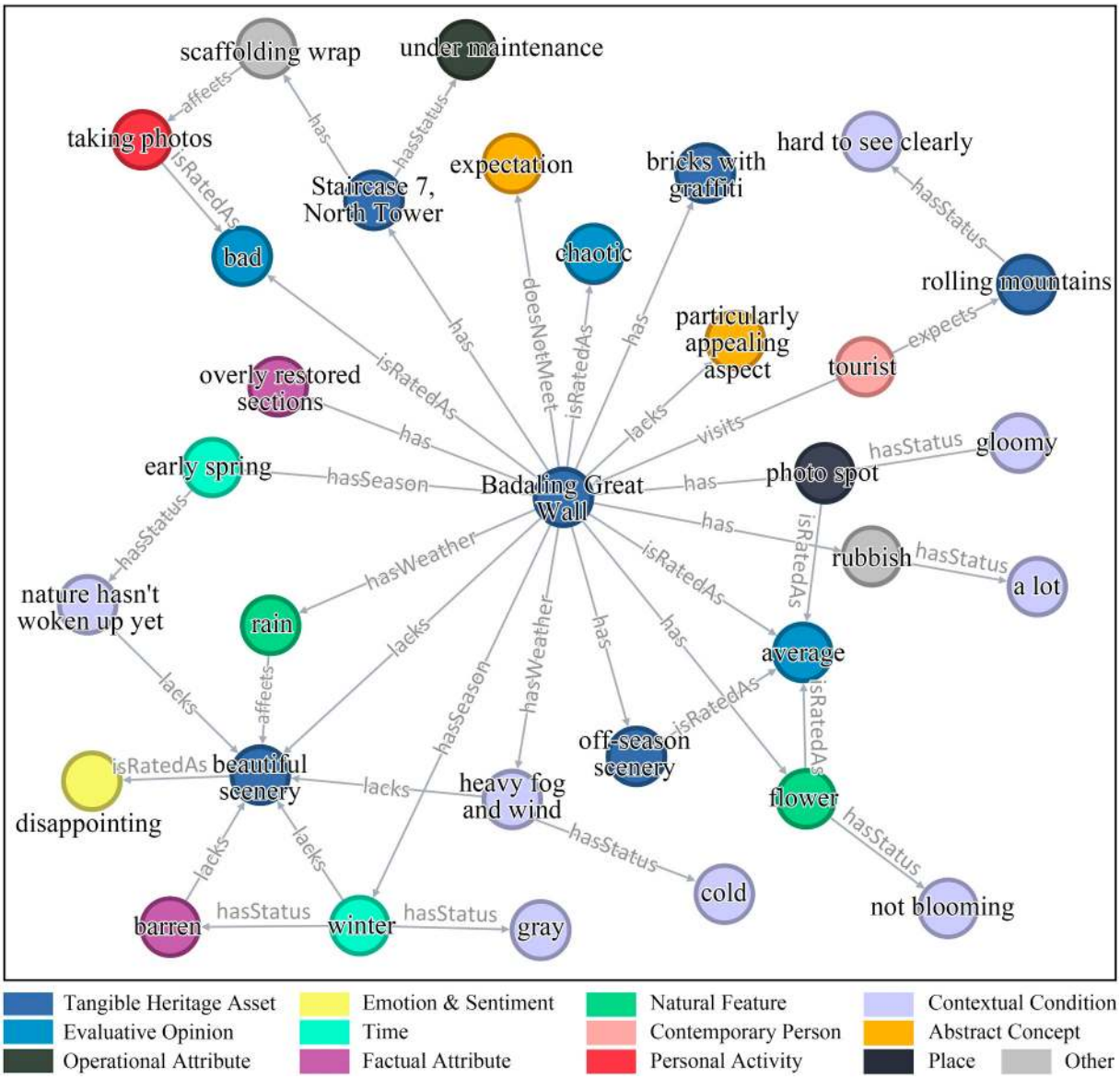


Fig. 11. A detailed view of entities and relations within a CHPG, capturing specific negative aesthetical perceptions.

stakeholder workshops. This mixed-methods approach can ensure a balanced strategy that respects the site's cultural significance while serving the needs of diverse user groups (Bai, 2023; Bai et al., 2023; Foroughi et al., 2023; Ginzarly et al., 2019; Silva & Roders, 2012).

Fourth, although our results suggest that factors such as regional context and heritage type play an important role in shaping perceived cultural values, these influences were not explicitly modeled in the present study. Future research could extend our workflow with more explicit typological and regional comparative designs, such as stratified analyses by heritage category and level, region, and historical function, or multilevel models that incorporate attraction-level attributes and city-level contextual factors. Such analyses would help clarify how public value perception is jointly shaped by visitor discourse, heritage characteristics, and broader socio-spatial context. Finally, the capabilities of our workflow are intrinsically linked to the ability of current LLMs, which still face challenges such as inherent biases (Luo et al., 2025; Zhao, Huang, Yang, et al., 2025), high inference time (Wang, Ma, et al., 2024; Yan et al., 2024), and high cost. As LLMs continue to evolve, future iterations of multi-agent LLM workflows will undoubtedly become more efficient, accurate, and powerful.

6. Conclusions

Methodologically, this study introduces a multi-agent LLM workflow to systematically identify public-perceived cultural values and extract, structure, and operationalize the supporting evidence. It assigns perception scores across eight cultural values and extracts value-related triples to construct CHPGs, which power a RAG system enabling stakeholders to query public feedback in plain language for evidence-based decision-making. To support future work, we also contributed public resources, including a custom ontology, two triple normalization dictionaries, and an entity typing dictionary.

Empirically, our large-scale analysis of 703 heritage attractions across 141 Chinese cities revealed how cultural values are expressed, distributed, and interrelated nationwide. In social media discourse, aesthetical and social values dominate, emphasizing personal experience and social connections, while economic discussions focus on direct-use aspects like ticket pricing. Historic and scientific values center on narratives and craftsmanship, respectively, while political and ecological values, though infrequent, convey abstract concepts of national identity and natural harmony. Spatially, aesthetical and social values distribute widely and cluster in Eastern, Southwestern, and Northern

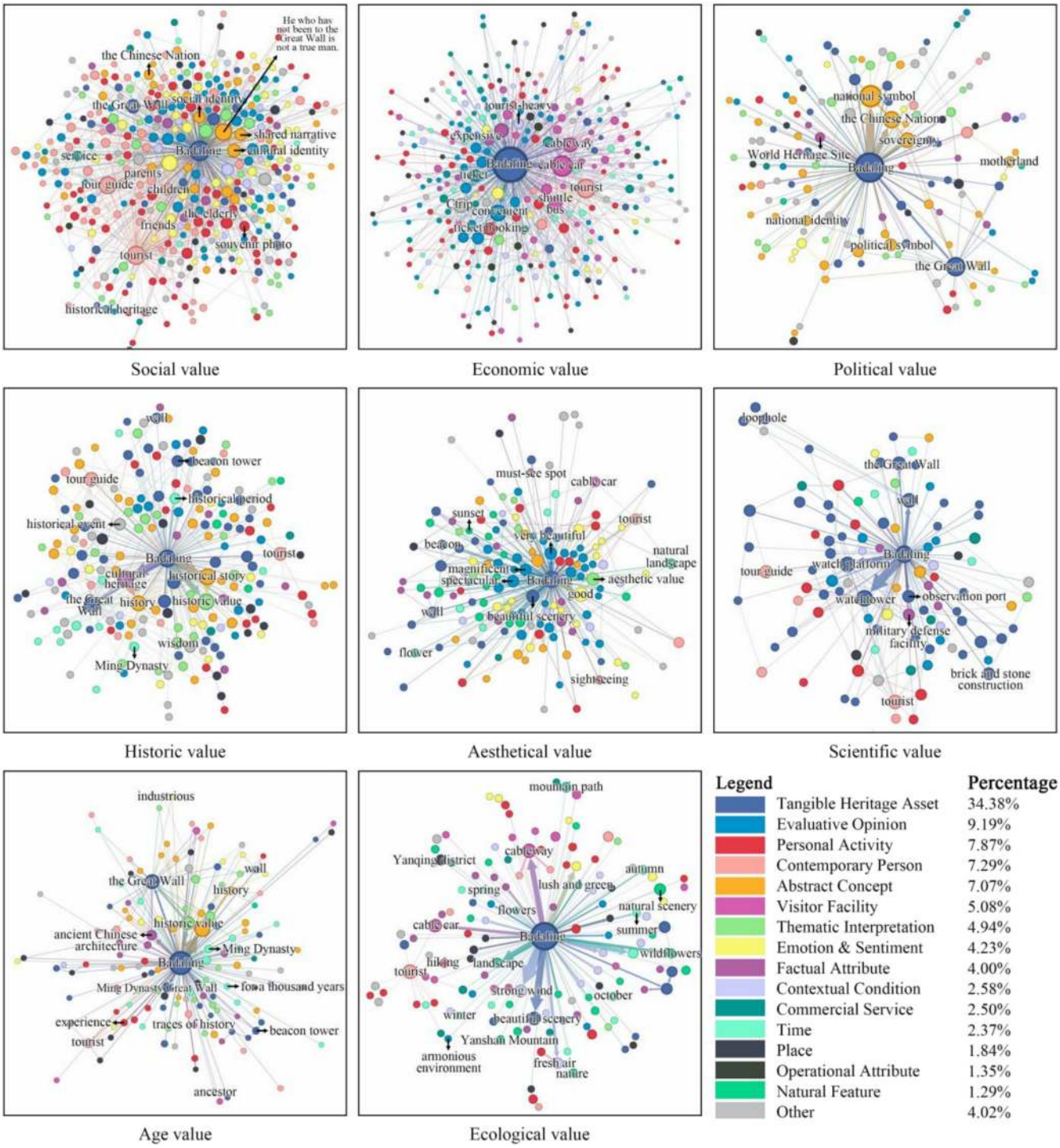


Fig. 12. CHPGs of the Badaling Great Wall across all eight values, showing variations in complexity and focus (frequency $\geq$ 4).

China. Economic value concentrates in prosperous metropolitan areas. Historic and age values cluster in ancient capitals, while scientific value appears selectively at technically remarkable sites. Political and ecological values show the most distinct concentrations, tied to revolutionary history sites and naturally harmonious settings respectively. Value interrelationships reveal that most dimensions positively correlate except ecological value, which stands distinct and negatively correlates with scientific value. The relationship between value perception and urban development produces four city profiles: High-Score/High-GDP cities with self-reinforcing cultural-economic cycles; High-Score/Low-GDP cities with untapped cultural assets; and two low-score categories representing missed opportunities or dual challenges. These patterns provide an evidence-based framework for differentiated heritage management strategies tailored to specific value profiles and urban contexts.

While this study offers a scalable, data-driven workflow, future research should conduct longitudinal analyses, incorporate multimodal data, and integrate these computational findings with traditional methods for a more holistic understanding.

CRediT authorship contribution statement

Xukai Zhao: Writing – original draft, Visualization, Software,

Methodology, Formal analysis, Data curation, Conceptualization. **Yuxing Lu:** Writing – original draft, Validation, Resources, Formal analysis, Data curation, Conceptualization. **He Huang:** Writing – review & editing, Validation, Funding acquisition, Conceptualization. **Guanhong Li:** Writing – original draft, Validation, Methodology, Conceptualization. **Chenxi Wang:** Validation, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cities.2026.107232.

Data availability

Data will be made available on request.

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